Aubree Thompson

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Summary

Analytical honors mathematics student with over a year of software engineering internship experience on a derivatives and investment team. Strong background in analytical thinking and modeling, with an ability to identify structure and insight in abstract problems. Recipient of many mathematics scholarships and academic honors, and a member of the mathematics honor society Kappa Mu Epsilon. Interested in opportunities involving quantitative finance, data-driven problem-solving, or research with dynamic systems or complex variables.

Education

Washburn University, Topeka, KS **May 2026**

Bachelor of Science in Mathematics | Concentration: Applied Statistics | Minor: Computer Information Science | GPA: 3.94
Relevant Coursework: Regression Analysis, Time Series Analysis, Mathematical Statistics I-II, Stochastic Processes, Statistical Computing, Data Mining, Database Management Systems, Linear Algebra

Honors & Awards

- President's List since Fall 2022
- Graduated Summa Cum Laude
- Served as the Kansas Delta Chapter representative on the 2026 Kappa Mu Epsilon Regional Convention Award Committee
- Mathematics Scholarship Recipient (Arden L. Kanode, Edward L. & Julie L. Glotzbach, John Ross & Lydia Greene Kinzer)

Professional Experience

Software Engineer Intern **March 2025 - Present**

Eldridge (Originally Security Benefit), Topeka, KS

- Assisted the investment and derivatives front office tech team of 7 software engineers, mainly working on Python backend development.
- Maintained and improved several data loaders and plugins in internal applications that are essential to the derivatives team's daily workflow.
- Reduced an internal application's data loading runtime by ~87% with vectorization and threading.
- Wrote SQL queries to validate data loads after code changes, investigate issues, and ensure the accuracy and completeness of the loaded data.
- Enhanced automated email outputs tied to data loaders, improving message clarity, structure, and usability for downstream data users.
- Created detailed change tickets and merge requests documenting needed file modifications, SQL for implementation, testing, and rollback procedures, while coordinating with the data team to support accurate and organized deployments.
- Standardized cross-OS compatibility for unit tests with efficient mocking and dynamic assertions, resolving environment-specific test failures between Windows workstations and Linux CI/CD pipelines.
- Migrated a legacy report generation system to a Python and C#/.NET architecture by automating API requests with JSON configurations and validating C# endpoints through Swagger. Created a Docker Compose setup for the existing containerized application to support bind mounts for local and network-shared drives, ensuring generated reports were uploaded to persistent shared storage rather than the container's temporary file system.
- Developed a centralized team resource homepage using Angular, HTML, SCSS/CSS, and Node.js. Created a draggable card-based UI that allowed users to organize frequently used business applications, internal tools, files, and resources, with persistent cookie storage to retain user preferences. Integrated a custom Azure Windows authentication package into the application, making the necessary code changes to support Windows-authenticated access.

Academic Projects

Comparison of Prediction Models for Stock Prices - Python, Machine Learning, Deep Learning, Time Series

- Recreated and extended a stock prediction study by comparing Holt-Winters, ARIMA, MARS, and LSTM models across historical stock data.
- Cleaned date and monetary fields, created train/test splits, evaluated performance with RMSE, and analyzed how COVID-era market disruption affected forecasting accuracy.

University Database - SQL, MySQL, ER Modeling, Normalization

- Designed and implemented a relational university database from scenario analysis through implementation, including ER modeling, normalized schema design, keys, constraints, and many-to-many relationship resolution.
- Wrote SQL queries, views, and triggers to support GPA risk checks, section codes, instructor load summaries, schedule conflicts, classroom usage, and course capacity rules.

Home Sale Price Regression Analysis - R, Regression, ggplot2, Statistical Modeling

- Built home sale price prediction models using exploratory analysis, dummy variables, variable screening, stepwise selection, interaction terms, residual diagnostics, and actual-versus-predicted visualization.

Ice Cream Production Time Series Analysis - R, Forecasting, SARIMA, Decomposition

- Analyzed monthly production data using decomposition, seasonality and autocorrelation diagnostics, Holt-Winters methods, harmonic regression, ARMA residual modeling, and SARIMA forecasting.

Technical Skills

Programming: Python, R, Java, C#, and SQL
Libraries: Pandas, NumPy, Matplotlib, ggplot2, tidyverse, and dplyr
Data & Statistics: Data cleaning, statistical modeling, regression analysis, time series analysis, hypothesis testing, machine learning, deep learning, supervised and unsupervised learning, model evaluation
Software: Microsoft Suite, Slack, VS Code, Visual Studio, GitLab, GitHub, Jira, Jira Change Management, Postman, Swagger, Docker Desktop, and RStudio
Core Strengths: Backend workflow development, database interaction, workflow optimization, analytical problem-solving, and technical communication